

FO fallback also extends
to all shared faces and edges

RK Stage 1

RK Stage 2

$$\mathbf{S}_c^{[n]}[i, j, k]$$

cache FO flux
and FO EMFs at $t^{[n]}$

compute HO flux
and HO EMFs at $t^{[n]}$

is the
proposed $\mathbf{S}_c^{[n+1/2]}[i, j, k]$
from HO calculations at $t^{[n]}$
valid?

no

yes

compute
 $\mathbf{S}_c^{[n+1/2]}[i, j, k]$ from
FO calculations at $t^{[n]}$

compute
 $\mathbf{S}_c^{[n+1/2]}[i, j, k]$ from
HO calculations at $t^{[n]}$

$$\mathbf{S}_c^{[n+1/2]}[i, j, k]$$

compute HO flux
and HO EMFs at $t^{[n+1/2]}$

is the
proposed $\mathbf{S}_c^{[n+1]}[i, j, k]$
from HO calculations at $t^{[n+1/2]}$
valid?

no

yes

compute
 $\mathbf{S}_c^{[n+1]}[i, j, k]$ from
FO calculations at $t^{[n]}$

compute
 $\mathbf{S}_c^{[n+1]}[i, j, k]$ from
HO calculations at $t^{[n+1/2]}$

$$\mathbf{S}_c^{[n+1]}[i, j, k]$$